

```
R1
Router>
Router>
Router>
Router>en
Router#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              10.0.0.1        YES manual up          up
Ethernet0/1              1.0.0.1         YES manual up          up
Ethernet0/2              unassigned      YES unset  administratively down down
Ethernet0/3              unassigned      YES unset  administratively down down
Router#sh route
% Ambiguous command: "sh route"
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PF8

Gateway of last resort is not set

  1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       1.0.0.0/8 is directly connected, Ethernet0/1
L       1.0.0.1/32 is directly connected, Ethernet0/1
S       2.0.0.0/8 [1/0] via 1.0.0.2
S       3.0.0.0/8 [1/0] via 1.0.0.2
S       4.0.0.0/8 [1/0] via 1.0.0.2
S       5.0.0.0/8 [1/0] via 1.0.0.2
S       6.0.0.0/8 [1/0] via 1.0.0.2
S       10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       135.82.0.0/16 [1/0] via 1.0.0.2
S       153.60.0.0/16 [1/0] via 1.0.0.2
S       182.46.0.0/16 [1/0] via 1.0.0.2
S       200.10.156.0/24 [1/0] via 1.0.0.2
Router#
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R2
ole
R2#
R2#
R2#sh ip int br
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 200.10.156.1 YES manual up
Ethernet0/1 2.0.0.1 YES manual up
Ethernet0/2 1.0.0.2 YES manual up
Ethernet0/3 6.0.0.1 YES manual up
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        +- replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    1.0.0.0/8 is directly connected, Ethernet0/2
L    1.0.0.2/32 is directly connected, Ethernet0/2
2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    2.0.0.0/8 is directly connected, Ethernet0/1
L    2.0.0.1/32 is directly connected, Ethernet0/1
S    3.0.0.0/8 [1/0] via 2.0.0.2
S    4.0.0.0/8 [1/0] via 2.0.0.2
S    5.0.0.0/8 [1/0] via 2.0.0.2
6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    6.0.0.0/8 is directly connected, Ethernet0/3
L    6.0.0.1/32 is directly connected, Ethernet0/3
S    10.0.0.0/8 [1/0] via 1.0.0.1
S    135.82.0.0/16 [1/0] via 2.0.0.2
S    153.60.0.0/16 [1/0] via 6.0.0.2
S    182.46.0.0/16 [1/0] via 2.0.0.2
200.10.156.0/24 is variably subnetted, 2 subnets, 2 masks
C    200.10.156.0/24 is directly connected, Ethernet0/0
L    200.10.156.1/32 is directly connected, Ethernet0/0
R2#
R2#
R2#
```

```
R3#
R3#
R3#
R3#sh ip int br
Interface                IP-Address      OK? Method Status    Protocol
Ethernet0/0              2.0.0.2         YES manual up        up
Ethernet0/1              3.0.0.1         YES manual up        up
Ethernet0/2              4.0.0.1         YES manual up        up
Ethernet0/3              unassigned      YES TFTP  up        up
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S  1.0.0.0/8 [1/0] via 2.0.0.1
S  2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
   C  2.0.0.0/8 is directly connected, Ethernet0/0
   L  2.0.0.1/32 is directly connected, Ethernet0/0
S  3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
   C  3.0.0.0/8 is directly connected, Ethernet0/1
   L  3.0.0.1/32 is directly connected, Ethernet0/1
S  4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
   C  4.0.0.0/8 is directly connected, Ethernet0/2
   L  4.0.0.1/32 is directly connected, Ethernet0/2
S  5.0.0.0/8 [1/0] via 4.0.0.2
S  6.0.0.0/8 [1/0] via 2.0.0.1
S  10.0.0.0/8 [1/0] via 2.0.0.1
S  135.82.0.0/16 [1/0] via 4.0.0.2
S  153.69.0.0/16 [1/0] via 2.0.0.1
S  182.46.0.0/16 [1/0] via 3.0.0.2
S  200.10.156.0/24 [1/0] via 1.0.0.1
R3#
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R3#
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R4
R4#sh ip int br
Interface      IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0    182.46.0.1      YES manual up          up
Ethernet0/1    3.0.0.2         YES manual up          up
Ethernet0/2    unassigned      YES TFTP  up          up
Ethernet0/3    unassigned      YES TFTP  up          up

R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S 1.0.0.0/8 [1/0] via 3.0.0.1
S 2.0.0.0/8 [1/0] via 3.0.0.1
S 3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   3.0.0.0/8 is directly connected, Ethernet0/1
L   3.0.0.2/32 is directly connected, Ethernet0/1
S 4.0.0.0/8 [1/0] via 3.0.0.1
S 5.0.0.0/8 [1/0] via 3.0.0.1
S 6.0.0.0/8 [1/0] via 3.0.0.1
S 10.0.0.0/8 [1/0] via 3.0.0.1
S 135.82.0.0/16 [1/0] via 3.0.0.1
S 153.69.0.0/16 [1/0] via 3.0.0.1
S 182.46.0.0/16 is variably subnetted, 2 subnets, 2 masks
C   182.46.0.0/16 is directly connected, Ethernet0/0
L   182.46.0.1/32 is directly connected, Ethernet0/0
S 200.10.150.0/24 [1/0] via 3.0.0.1

R4#
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R5
R5#sh ip int br
Interface          IP-Address      OK? Method Status      Protocol
Ethernet0/0        5.0.0.1         YES manual up          up
Ethernet0/1        4.0.0.2         YES manual up          up
Ethernet0/2        unassigned      YES TFTP  up          up
Ethernet0/3        unassigned      YES TFTP  up          up
R5#
R5#
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 4.0.0.1
S    2.0.0.0/8 [1/0] via 4.0.0.1
S    3.0.0.0/8 [1/0] via 4.0.0.1
C    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
     C    4.0.0.0/8 is directly connected, Ethernet0/1
     L    4.0.0.2/32 is directly connected, Ethernet0/1
C    5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
     C    5.0.0.0/8 is directly connected, Ethernet0/0
     L    5.0.0.1/32 is directly connected, Ethernet0/0
S    6.0.0.0/8 [1/0] via 4.0.0.1
S    10.0.0.0/8 [1/0] via 4.0.0.1
S    135.82.0.0/15 [1/0] via 5.0.0.2
S    153.69.0.0/15 [1/0] via 4.0.0.1
S    182.46.0.0/15 [1/0] via 4.0.0.1
S    200.10.156.0/24 [1/0] via 4.0.0.1
R5#
R5#
R5#
R5#
```

```
R6#sh ip int br
Interface          IP-Address      OK? Method Status      Protocol
Ethernet0/0        135.82.0.1      YES manual up          up
Ethernet0/1        5.0.0.2         YES manual up          up
Ethernet0/2        unassigned      YES TFTP  up          up
Ethernet0/3        unassigned      YES TFTP  up          up

R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 5.0.0.1
S    2.0.0.0/8 [1/0] via 5.0.0.1
S    3.0.0.0/8 [1/0] via 5.0.0.1
S    4.0.0.0/8 [1/0] via 5.0.0.1
S    5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       5.0.0.0/8 is directly connected, Ethernet0/1
L       5.0.0.2/32 is directly connected, Ethernet0/1
S    6.0.0.0/8 [1/0] via 5.0.0.1
S   10.0.0.0/8 [1/0] via 5.0.0.1
S  135.82.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     135.82.0.0/16 is directly connected, Ethernet0/0
L     135.82.0.1/32 is directly connected, Ethernet0/0
S  153.69.0.0/16 [1/0] via 5.0.0.1
S  182.46.0.0/16 [1/0] via 5.0.0.1
S  200.10.156.0/24 [1/0] via 5.0.0.1

R6#
R6#
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R7#
R7#
R7#sh ip int br
Interface              IP-Address      OK? Method Status      Prot
-----
GigabitEthernet0/0     153.69.0.1      YES manual up          up
GigabitEthernet0/1     6.0.0.2         YES manual up          up
GigabitEthernet0/2     unassigned      YES TFTP  up          up
GigabitEthernet0/3     unassigned      YES TFTP  up          up

R7#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 6.0.0.1
S    2.0.0.0/8 [1/0] via 6.0.0.1
S    3.0.0.0/8 [1/0] via 6.0.0.1
S    4.0.0.0/8 [1/0] via 6.0.0.1
S    5.0.0.0/8 [1/0] via 6.0.0.1
    6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        6.0.0.0/15 is directly connected, Ethernet0/1
L        6.0.0.2/32 is directly connected, Ethernet0/1
S    10.0.0.0/8 [1/0] via 6.0.0.1
S    135.82.0.0/16 [1/0] via 6.0.0.1
    153.69.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        153.69.0.0/16 is directly connected, Ethernet0/0
L        153.69.0.1/32 is directly connected, Ethernet0/0
S    182.46.0.0/16 [1/0] via 6.0.0.1
S    200.10.156.0/24 [1/0] via 6.0.0.1

R7#
R7#
R7#
R7#
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